# BASIC STATISTICS

**Descriptive Analytics and Data Preprocessing on Sales & Discounts Dataset**

## *Introduction*

- To perform descriptive analytics, visualize data distributions, and preprocess the dataset for further analysis.

## *Descriptive Analytics for Numerical Columns*

- Objective: To compute and analyze basic statistical measures for numerical columns in the dataset.
- Steps:

  Load the dataset into a data analysis tool or programming environment (e.g., Python with pandas library).
  Identify numerical columns in the dataset.
  Calculate the mean, median, mode, and standard deviation for these columns.
  Provide a brief interpretation of these statistics.

## *Data Visualization*

- **Objective**: To visualize the distribution and relationship of numerical and categorical variables in the dataset.
- **Histograms**:

  Plot histograms for each numerical column.
  Analyze the distribution (e.g., skewness, presence of outliers) and provide inferences.
- **Box Plots**:

  Create boxplots for numerical variables to identify outliers and the interquartile range.
  Discuss any findings, such as extreme values or unusual distributions.
- **Bar Chart Analysis for Categorical Column:**

  Identify categorical columns in the dataset.
  Create bar charts to visualize the frequency or count of each category.
  Analyze the distribution of categories and provide insights.

## *Conclusion*

- Summarize the key findings from the descriptive analytics and data visualizations.